Politecnico di Milano
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Atom Optics and Quantum Technologies
July 15th, 2025

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Part 1 - Dipole force.

In the semiclassical two-level atom picture, laser light can exert two distinct types of forces on atoms, the resonance-radiation pressure force and the dipole force:

$$\vec{F}_{RRP}(\vec{r}) = -\frac{|\mu_{12}|^2}{2\hbar}\frac{\Delta\omega\, E_0(\vec{r})\cdot\nabla E_0(\vec{r})}{\Delta\omega^2 + \left(\frac{\Gamma}{2}\right)^2 + \frac{|\Omega_{12}|^2}{2}}$$

$$\vec{F}_{dip}(\vec{r}) = -\frac{|\mu_{12}|^2}{2\hbar}\left(\frac{\Gamma}{2}\right)\frac{{E_0}^2(\vec{r})\cdot\nabla\varphi(\vec{r})}{\Delta\omega^2 + \left(\frac{\Gamma}{2}\right)^2 + \frac{|\Omega_{12}|^2}{2}}$$

a) Explain the mathematical link between the dipole force the gradient of the intensity of the em field associated to the laser.
b) Provide a qualitative explanation of the mechanism by which the dipole force arises from the interaction between the atom's induced dipole and the spatial gradient of the laser field.
c) Derive the expression of the potential energy well $U_{dip}(\vec{r})$ associated to the dipole force and draw a qualitative plot of it ($k = 1.38 \times 10^{-23}$ J·K^{-1}).
d) Assume that you place a dipole-force-based atomic trap at exit of an oven evaporating sodium atoms. Due to experimental conditions, the potential well has a depth of $U_0 \cong 10^{-25}$ J. Compute the characteristic temperature T_{trap} below which atoms can be trapped.
e) Discuss, under proper approximations, what happens when both the resonance-radiation-pressure force and dipole force act togheter, and show under what circumstances one dominates over the other.

Part 2 – Ramsey fringes theory

Consider a two-level atomic system subjected to two coherent microwave (or optical) pulses of duration τ, separated by a free evolution time T.

a) Derive the expression for the transition probability of the atom. Explain how the Ramsey method allows for improved spectral resolution compared to a single continuous pulse, and why this is advantageous in the context of atomic clocks.
b) Draw a possible experimental setup for a Ramsey cavity (e.g., with two interaction zones and a free-flight region). Explain how the interference fringes obtained in the transition probability result from the phase difference accumulated during the free evolution time T.

Part 3 – Optical molasses

Derive the expression of the net force acting within a 3D optical molosses and show that it is ultimately equivalent to a friction force.